C K Pithawala College of Engineering & Technology
Computer Engineering Department

Subject: Artificial Intelligence (3170716)
Year (Semester): BE - IV (VII)

Course-Integrated Practical & Analytical Task (CIPAT)

Details of Evaluation :

Teaching and Examination Scheme:

Teaching Scheme			Credits	Examination Marks				Total Marks
				Theory Marks		Practical Marks		
L	T	P	C	ESE (E)	PA (M)	ESE (V)	PA (I)	
3	0	2	4	70	30	30	20	150

1. Breakup of CIPAT [Marks - 40] :

A) Attendance Component [10 Marks]
B) Problem Solving Component [30 Marks]

A) Attendance [Marks - 10] :

- Minimum attendance of 75 % is mandatory, as per GTU Norms. Attendance Rubrics has been explained below:

Attendance (%)	Marks (10)
>90 %	10
70 - 80 %	8
50 - 75 %	5
< 50 %	2

B) Problem Solving Component [Marks -30] :

Problem Statement-1 : Develop an AI-based Expert System to solve a Real-World problem [Marks-15]

System should:

- Identify the problem and define the objective.
- Represent the problem using suitable facts and rules.
- Apply an appropriate AI search technique to solve the problem.
- Implement the solution using Prolog.
- Study and explain how modern AI technologies solve similar problems in real-world applications.
- Demonstrate the working of the system using sample queries and outputs.

Example :

1. Smart Hospital Diagnosis Assistant: Develop a Prolog-based Expert system that helps diagnose common diseases based on patient symptoms and suggests appropriate medical advice.

- Apply an AI search technique to identify possible diseases.
- Represent symptoms, diseases, and treatments using Prolog facts and rules.
- Study modern AI applications in healthcare such as medical diagnosis and predictive analytics.
- Implement and demonstrate the expert system using Prolog.

2. Job Recommendation System: Develop an expert system that recommends suitable jobs based on educational qualifications, skills, experience, and interests.

- Apply search techniques to identify suitable job opportunities.
- Represent candidate profiles and job requirements using Prolog.
- Study AI-based recruitment platforms and resume screening systems.
- Develop the recommendation system using Prolog.

3. Smart Travel Planner: Design an AI-based travel planning system that suggests the best travel route considering distance, cost, and travel time.

- Apply search algorithms for route planning.
- Represent cities, routes, and travel constraints using Prolog.

- Study AI applications used in navigation systems like Google Maps.
- Implement the planner in Prolog.

4. Restaurant Recommendation System: Develop a system that recommends restaurants based on cuisine, budget, location, and customer preferences.

- Search suitable restaurants based on user requirements.
- Represent restaurant information using Prolog facts and rules.
- Study AI recommendation systems used in food delivery applications.
- Implement the recommendation engine using Prolog.

5. Student Course Advisor: Develop an expert system that recommends elective courses based on students' interests, academic performance, and career goals.

- Apply search techniques to identify suitable courses.
- Represent course eligibility and recommendation rules in Prolog.
- Study AI-based personalized learning systems.
- Develop the advisor using Prolog.

6. Smart Agriculture Advisor: Develop an intelligent system that recommends suitable crops based on soil type, rainfall, climate, and season.

- Apply search techniques to identify suitable crops.
- Represent agricultural knowledge using Prolog.
- Study AI applications in precision agriculture.
- Implement the expert system in Prolog.

7. Library Book Recommendation System: Develop a system that recommends books based on user interests, previous reading history, and preferred categories.

- Apply search techniques for book selection.
- Represent books and recommendation rules using Prolog.
- Study AI recommendation systems used in digital libraries.
- Implement the recommendation system using Prolog.

8. Banking Loan Approval Advisor: Develop an expert system that assists banks in evaluating loan eligibility based on customer information.

- Apply AI search to evaluate loan applications.
- Represent banking rules using Prolog.
- Study AI applications in banking and financial services.

- Implement the system using Prolog.

9. Vehicle Service Recommendation System: Develop an expert system that recommends vehicle maintenance services based on vehicle condition and usage.

- Search suitable maintenance solutions.
- Represent maintenance knowledge using Prolog.
- Study AI applications in predictive vehicle maintenance.
- Implement the system in Prolog.

10. Online Shopping Recommendation System: Develop a recommendation system that suggests products based on user preferences and purchase history.

- Search suitable products using AI techniques.
- Represent products and customer preferences using Prolog.
- Study AI recommendation systems used in e-commerce.
- Implement the system using Prolog.

11. Legal Advisory Expert System: Develop an expert system that provides basic legal guidance based on user queries and predefined legal rules.

- Search legal solutions based on user cases.
- Represent legal rules using Prolog.
- Study AI applications in legal technology.
- Implement the advisory system using Prolog.

12. Criminal Investigation Assistant: Develop a system that identifies possible suspects based on available evidence and investigation rules.

- Apply AI search to identify suspects.
- Represent evidence and investigation rules using Prolog.
- Study AI applications in crime analysis and digital forensics.
- Implement the system using Prolog.

13. Disaster Management Advisor: Develop an expert system that provides safety recommendations during natural disasters based on disaster type and severity.

- Search suitable emergency actions.
- Represent disaster management rules using Prolog.
- Study AI applications in disaster prediction and emergency response.

- Implement the expert system using Prolog.

14. Animal Disease Diagnosis System: Develop a Prolog-based expert system that identifies common animal diseases based on symptoms and recommends preliminary treatment.

- Apply search techniques to identify diseases.
- Represent veterinary knowledge using Prolog.
- Study AI applications in veterinary healthcare.
- Develop the expert system using Prolog.

15. Smart Waste Management Advisor: Develop an intelligent system that recommends waste segregation and disposal methods based on different types of waste.

- Search appropriate waste management solutions.
- Represent waste categories and disposal rules using Prolog.
- Study AI applications in smart city waste management.
- Implement the recommendation system using Prolog.

CO-Wise Rubrics Mapping : (Total- 15 Marks)

CO	Evaluation Criteria	Marks
CO1	Problem analysis and search technique	3
CO2	Knowledge representation using facts and rules	3
CO4	Study of modern AI applications	2
CO5	Prolog implementation and demonstration	4
	Report	3
	Total	**15**

Problem Statement-1: AI_CIPAT Report Format Problem-1-2026-27

Problem Statement-2 : Game Implementation [Marks-15]

1. **Maze Escape Game:** Develop a two-player maze game in which players compete to reach a target location. The AI player should use a suitable search technique such as BFS, DFS, or A* to find possible paths and use Minimax or Alpha-Beta pruning to decide the best move against the opponent.

2. **Treasure Hunt Game:** Develop a two-player treasure hunt game in which players move through a grid to collect treasure. The AI player should use a search technique to explore the game environment and use Minimax or Alpha-Beta pruning to select the best move while considering the opponent's actions.

3. **Grid Strategy Game:** Develop a two-player grid-based strategy game where players compete to achieve a predefined goal. The AI player should explore possible game states using a suitable search technique and use Minimax or Alpha-Beta pruning to select the optimal move.

4. **Tic-Tac-Toe Strategy Game:** Develop an intelligent Tic-Tac-Toe game in which the computer plays against a human. The AI should explore possible game states and use the **Minimax algorithm** to select the best move. The implementation should show how the game tree is searched and how the best move is selected.

CO-Wise Rubrics Mapping : (Total- 15 Marks)

Co	Evaluation Criteria	Marks
CO1	Heuristic Function for individual problem statement	4
CO3	Implementation of the given gaming problem statement	6
CO3	Understanding and explaining the game implementation.	3
	Report	2
	Total	**15**

Problem Statement-2: AI_CIPAT Report Format Problem-2-2026-27

Important Notes:

- Make a group of **05** students from your **Practical Batch** only and Register two topics to the respective Lab faculty.
- All the groups are informed to have **Distinct topics**.
- A total of 30 marks will be mapped to 20 marks for the final examination.
- Provide the team details during your lab session to your **lab faculty** only.
- At the time of Evaluation the group must be present.
- Participation of all members during the submission and development of the system must be counted.
- Evaluation will be conducted strictly during scheduled lab hours only. Any requests or submissions after lab hours will not be entertained. Students must adhere to this rule without exception.
- **Prepare two distinct reports for two different problem statements.**

Practical Internal:PA [20 Marks]

Sr. No	Rubrics	Marks
1	Implementation	10
2	File	5
3	Viva	5

Course Outcomes:

COs (Course Outcomes)	COs (Course Outcomes)	Marks % Weightage
CO1	Understand the search technique procedures applied to real world problems	25
CO2	Understand and use various types of logic and knowledge representation schemes.	30
CO3	Understand various Game Playing techniques and apply them in programs.	15
CO4	Gain knowledge in AI Applications and advances in Artificial Intelligence	20
CO5	Use Prolog Programming language using predicate logic	10

Subject Coordinator

Prof. Chaitali Sheth

Subject Faculties

Prof. Kushal Patel